

Francisco Castro

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Professional Summary

Results-driven scientist and technology professional and manager with extensive experience in human-centered design, data, and AI application development across industry and academic settings worldwide. Proven track record of securing \$1M+ in research funding, publishing award-winning research articles, and delivering impactful technology solutions at international organizations. Skilled at leading and collaborating in cross-functional, multi-institutional teams to build innovative technology products that reach users globally.

Experience

New York University, Research Scientist - Human-Centered Technologies and AI

New York, NY
2022 – Present

- Successfully acquired grant awards from government and private institutions for computing, technology, AI, and education research, totaling \$1M+ USD
- Published and presented 40+ research articles in high-quality, peer-reviewed international conferences and journals from multi-institutional collaborative projects
- Led and managed cross-functional, multi-institutional, and international project teams for research, technology development, and program implementation — projects, tools, and platforms researched/developed are at beacons.ai/franciscocastro/projects 🔗

Accenture, Senior Data and AI Advanced Application Engineering Analyst

Boston, MA
2020 - 2021

- Android Developer, Global Media and Technology – Developed and integrated an AI assistant customer experience with multi-language support using Kotlin into a customer subscription Android application
- Product Owner, Multinational Consumer Electronics – Collaborated with cross-functional teams and stakeholders to manage product backlogs for the company's technology transfer into a cloud-based subscriptions management platform
- Represented the company in national conferences as part of the Accenture Employee Resource Groups

Amadeus, Human-Computer Interaction Research and UX Engineer Intern

Waltham, MA
2018

- Lead developer of an online airline operations response system that integrated interactive visualizations of flight information, airline delays, and situational response protocols with D3.js, Node.js, and Angular
- Conducted user research to analyze the behaviors and system interactions of airline operations managers

Brown University, Visiting Research Scientist

Providence, RI
2017 - 2020

- Conducted human factors research on software design systems and code design processes across multiple project sites in Boston MA, Worcester MA, and Providence RI
- Published and presented 15+ research articles, including award-winning research in high-quality, peer-reviewed international conferences

DataCamp, Data Science Content Development Intern

Cambridge, MA
2016

- Designed and deployed online interactive data science courses used by about 1.3 million users worldwide to date — see courses at beacons.ai/franciscocastro/projects 🔗
- Developed software to automate the evaluation and feedback-delivery of user submissions

Nara Institute of Science and Technology, Visiting Research Scientist

Nara, Japan
2013

- Developed the automatic speech recognition system of the Augmented Human Commu-

nication Laboratory by mining web and video resources speech patterns using in-house Python tools and libraries

Ateneo de Manila University, UX Designer and Researcher

Manila, Philippines

2013 - 2014

- Deployed the ehealth.ph Android mobile health information system (eHealth TABLET) to local government implementation sites across five regions in the Philippines
- Trained medical and public health professionals on mobile health system usage and health information and care-delivery digitalization
- Designed Android and web user interfaces for a traffic and air pollution analysis system and conducted usability testing to enhance the system user experience

Education

PhD	Worcester Polytechnic Institute , Computer Science	2020
MS	Ateneo de Manila University , Information Technology	2014
BS	University of the Philippines Los Baños , Computer Science	2010

Selected Publications

XAIxArts Manifesto: Explainable AI for the Arts	2025
ACM CHI Conference on Human Factors in Computing Systems — doi.org/10.1145/3706599.3716227	
Body & Code: Distributed Cognition Exploration Into Dance & Computing Learning	2024
ACM Conference on Creativity & Cognition — doi.org/10.1145/3635636.3656206	
Time-on-Task Metrics for Predicting Performance, <i>Best Paper Award</i>	2022
ACM Technical Symposium on Computer Science Education — doi.org/10.1145/3478431.3499359	
Designing a Multi-faceted SOLO Taxonomy to Track Program Design Skills Through an Entire Course, <i>Best Paper Award</i>	2017
ACM Koli Calling International Conference on Computing Education Research — doi.org/10.1145/3141880.3141891	
• <i>More publications:</i> Google Scholar — scholar.google.com/citations?user=ubFX8kwAAAAJ , ORCID — orcid.org/0000-0002-3940-5334 , ACM Digital Library — dl.acm.org/profile/99658729217	

Skills, Tools, and Technologies

Research Tools and Techniques: Experimental Design, Qualitative/Quantitative/Mixed-Methods Research, Interviews, Focus Groups, Ethnography, Card Sorting, Survey Design and Platforms (Qualtrics, SurveyMonkey), Qualitative Data Analysis Software (NVivo, ATLAS.ti, Dedoose), Research Management Platforms (Zotero, Mendeley), Google Colab

Project Management and Productivity Platforms: GitHub Projects, Trello, Asana, Google Workspace, Microsoft 365, Zoom, Microsoft Teams, Slack, Miro

Software Development and Management: Python, JavaScript, Node.js, Java, Visual Studio Code, Git, GitHub, Bitbucket, MySQL, PostgreSQL, MongoDB, DBA, Anaconda, HTML, CSS, Eleventy, Bootstrap

AI Tools: Claude, ChatGPT, NotebookLM, Gemini

Professional Organizations

Association for Computing Machinery (ACM)

- Special Interest Group on Computer-Human Interaction (SIGCHI)
- Special Interest Group on Computer Science Education (SIGCSE)
- ACM Athena Award Committee Member